Записки научных
семинаров ПОМИ
Том 1, 2026 г.

Arsenii Varaksin, Alexander Shiianov, David Sverdlov
Mentor: Serguei Barannikov (Skoltech)
Correspondence: varbarn22@gmail.com

WHAT THE LATENT TRAJECTORY OF A RECURRENT-DEPTH TRANSFORMER ENCODES, AND WHAT OUR MEASUREMENTS WERE ACTUALLY MEASURING

Abstract. A recurrent-depth transformer's latent trajectory is usually read as a trace of its reasoning. Measuring that reading directly on Huginn-3.5B, we find that the trajectory's most conspicuous geometric structure is selected by the surface form of the instruction rather than by the task, and that much of what we had scored as capability was answer formatting. A single instruction word switches the recurrent map between a settling regime and a damped period-6 rotation at an identical token count (36 of 36 paired cells), and the switch is causally controlled by one row of the embedding matrix. Scoring the first generated token reads 0.125 where the answer is present in 0.781 of the generated text, and depth raises the rate of opening with prose twentyfold while leaving the rate of opening with the answer flat. Supplying the output format reproduces the published benchmark number to within 4.1 points under the protocol its authors used. The two findings meet in a timing fact: the answer is fixed some twelve unrolls before the regime becomes measurable at all. We report the claims this project withdrew alongside those it kept.

§1. Introduction

A recurrent-depth transformer reuses one weight-tied block many times at inference, so the sequence of latent states it passes through is often read as a trace of reasoning — something whose shape should reflect the computation being performed. We tested that reading on Huginn-3.5B [1] and report two things.

The first is that the map has real, sharp, causally controllable structure, but that this structure is selected by the surface form of the instruction rather than by the task. The second is that a substantial part of our own

Key words and phrases: recurrent-depth transformers and latent reasoning and interpretability and evaluation protocols and dynamical systems.

measured capability record — and, we argue, of the standard way such models are scored — was measuring whether the model emits its answer in the position a scorer reads, not whether it has the answer.

We report both, together with the claims we withdrew in the process. The withdrawals are included deliberately: in several cases the instrument, not the model, produced the result, and a report that omitted them would misrepresent how reliable the remainder is.

It is worth saying at the outset what this report does not offer. Every number comes from one model, one checkpoint and one seeded initial state per condition; nothing here supports a claim about recurrent-depth models in general. The behavioural test of H3 is absent rather than negative. Our accuracy figures inherit an uncorrected surface-form bias from likelihood scoring, which the published figure we compare against also carries, so the comparison survives it while the absolute values of both remain questionable in the same direction. Several nulls rest on paired samples of twelve to twenty-four units, which exclude large effects and not small ones.

Concretely, this report contributes:

- a validated scalar instrument for latent rotation, with a threshold that reproduces the hand-assigned regime label on 691 of 696 orbits and 688 held out, replacing inspection of two-dimensional projections;
- the finding that this regime is selected by one instruction word at fixed token count and is causally controlled by a single row of the embedding matrix;
- a decomposition of measured capability into answer availability and answer formatting, which reproduces the model's published benchmark number under its authors' own protocol;
- a timing result placing the regime and the answer in non-overlapping windows of the same trajectory, which reinterprets our own behavioural nulls (Figure 1).

§2. Model and Protocol

Huginn-3.5B (tomg-group-umd/huginn-0125, revision bb6621b6...) embeds its input, applies a two-layer prelude, then iterates a four-layer core block r times, and finishes with a two-layer coda, a final normalisation and the language-model head, where the prelude output e is re-injected at every unroll through the adapter, which receives the concatenation of the

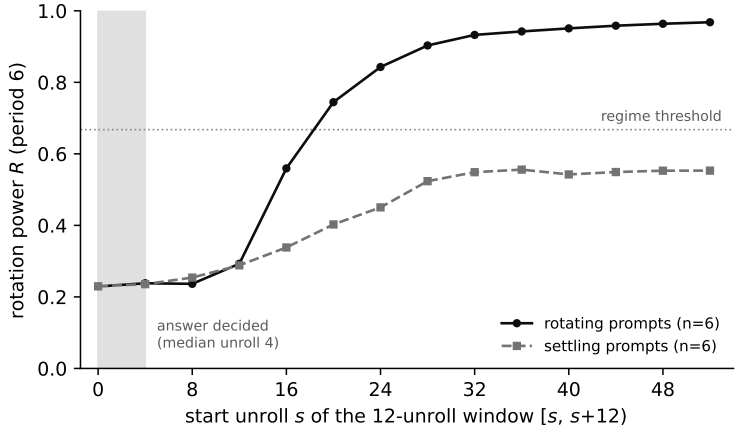

Figure 1. Rotation power in a sliding 12-unroll window, median over six rotating and six settling prompts. The axis is the window start: each point summarises unrolls $[s, s+12)$, twelve being the smallest window in which a period-6 orbit is defined at all. The two families are indistinguishable until the window beginning at unroll 16, by which point the answer has long been settled (shaded; median unroll 4). Higher R means more of the trajectory's power sits at period 6.

current state and e. Following the DEQ reading [6], we treat e as the map's parameter and the latent state h as its state; the released source supports this, since the block index is threaded through the unroll but never enters the block computation.

All measurements are inference-only in float32 and hook the output of core_block[-1]. Unless stated, depth is $r = 32$ or $r = 48$. Every forward is seeded, because the initial state is drawn from an unseeded generator by default.

§3. The Recurrent Map Has a Sharp, Causally Controlled Regime

3.1. Measuring rotation without projecting. The published account of this model's latent dynamics rests on projecting trajectories onto their leading principal components and judging the resulting pictures by eye; no period is stated, no null is computed, and no threshold separates an orbit from a non-orbit. We replace the picture with a scalar. Taking the discrete Fourier transform along the unroll axis, in the full 5280-dimensional state space and without any projection, we define the rotation power R as the fraction of non-DC power falling in the period-6 bin over a 24-unroll tail.

Three properties make it usable where a projection is not. It reproduces its own spectral definition to 4×10^{-8}. A single threshold, $R > 0.6677$, reproduces the hand-assigned binary label on 691 of 696 orbits and on 688 held out leave-one-bank-out. And rotating prompts peak at period 6 specifically — dominant bin exactly 6.00, power fraction 0.820 and 0.962 on the two strongest — rather than merely somewhere, so the statistic is not standing in for something broader.

The difference decides what can be claimed: a two-dimensional projection is fitted to the data it is used to judge, confounds damping with rotation, and supports neither a negative nor a threshold. Every claim below that a prompt does not rotate depends on having a statistic rather than a picture. Appendix E gives the argument in full, including an error of exactly this kind that we made ourselves and caught only because the statistic had a gate.

3.2. One instruction word switches the regime. Holding the sequence, the required answer and the token count fixed at 53, the instruction "report the largest symbol" produces a damped period-6 rotation in 36 of 36 paired cells, while "largest element" and "largest item" produce settling in 0 of 36. The selecting property is not semantic, not orthographic, and not tokenisation: the semantic neighbours of symbol rotate in 0 of 144 orbits, those of element in 96 of 120, and form-matched controls chosen to denote nothing similar (symptom, cymbal, symmetry) rotate in 72 of 72.

A single threshold on a continuous rotation statistic reproduces the binary label on 691 of 696 orbits, and on 688 of 696 held out leave-one-bank-out.

3.3. The switch is causally controlled by one embedding row. Editing a single row of the embedding matrix, with the prompt string and every token identifier untouched, flips the regime. Both instrument gates are bit-exact: substituting the row for itself reproduces the unpatched orbit at 0.000e+00, and substituting the target's row reproduces the target's own orbit.

Scope. The within-regime control failed in that experiment and was only partly resolved afterwards: rescaling the interpolated row to constant norm rescues one noun pair and not another. The licensed statement is therefore that the embedding row causally determines the regime within a small, direction-dependent neighbourhood — not that the regime is a simple function of the embedding.

3.4. The regime belongs to the prompt, not to the position measured. Every measurement above reads one token position. Because the published figures for this model are drawn at interior positions, we re-measured the rotation statistic at every position. Prompts we label rotating rotate at 62.0% of their positions (range 0.614–0.623 over nine prompts); prompts we label settling rotate at 0.000 — no overlap and no intermediate case. Within a rotating prompt, rotation switches on at a single boundary and holds to the end.

3.5. No hysteresis, and the contraction rate predicts a timescale. Swapping e mid-trajectory and leaving it swapped, 62 of 64 switches take, at switch points from unroll 1 to unroll 48: the regime follows the parameter the map currently has rather than the trajectory's history. Entering rotation takes a median of 15.5 unrolls and leaving it 1.0 — a fifteenfold asymmetry measured by the identical detector. The entering time brackets a prediction registered before the run from the measured contraction rate $\rho \approx 0.83$, namely $\ln(0.05)/\ln(0.83) = 16.1$ unrolls.

Scope. The detector uses a sliding window that carries its own lag, so the measured relaxation is an upper bound. The asymmetry, measured identically in both directions, is the robust part.

§4. What the Capability Measurements Were Measuring

4.1. The answer is present but not in the scored position. Generating and reading the text, rather than reading the rank of the gold token at the first generated position, changes the picture. First-token exact match reads 0.125 where the answer is actually produced in 0.781 of generations, against a measured cross-item false-positive rate of 0.042–0.152. The token that displaces the gold is The: it opens 39.4% of 1260 banked generations. On tasks the model can do, the gold's rank at the final unroll never exceeds 35 across 2720 draws.

4.2. Depth increases prose, not answers. Across $r = 2, 4, 8, 16, 32$ the rate at which a generation opens with The runs $0.032 \to 0.099 \to 0.631 \to 0.560 \to 0.647$, while the rate at which it opens with the gold does not move ($p = 0.515$). More depth therefore makes the standard first-token metric worse while the underlying output improves.

4.3. Supplying the format recovers the published number, by two routes. The published ARC-Easy figure for this model is 0.699 at $r = 32$. Our reading of the released evaluation code identifies the protocol as zero-shot lm-eval-harness with normalised accuracy; that is our bare-prompt, no-option-list, character-normalised arm, which reads 0.658. We state this as a hypothesis about their protocol rather than as an established fact: it rests on our reading of their repository and the paper's table caption, not on a statement from the authors, and the arm it selects is not the closest of ours to their number. Independently, five in-context examples under option-letter argmax give 0.723, paired, with 35 items fixed against 4 broken (exact McNemar $p = 3.35\text{e}{-07}$).

4.4. Instruction fails where demonstration succeeds. Instructing the model to "reply with only the answer" yields no improvement over a bare prompt; asking it to place its answer in \boxed{} produced fewer such answers than not asking (2 of 36 against 4 of 36). Two in-context examples, by contrast, move oracle accuracy by +0.221 (exact McNemar $p = 9.3\text{e}{-09}$) and take three families from 0.000 to 1.000 at the final unroll.

Placing the identical instruction in the system turn rather than the user turn—the form another group used to suppress prose in this model—buys one item in twenty-four (paired, $p = 1.0$), so the failure is not a property of the turn. The one qualified exception does not fight the prose frame but appends to it: asking the model to work through the problem and then write Answer: on the last line raises parsed-exact accuracy from 0.000 to 0.167, gaining four items and losing none ($p = 0.125$, $n = 24$). Suggestive rather than established, and consistent with the prose being where the computation happens.

4.5. Steering the map's parameter reproduces neither effect. Because e is re-injected at every unroll, a vector added to it is applied r times rather than once. Across twelve surveyed papers on this model family we found no prior steering-vector or activation-addition experiment on a recurrent-depth model, so we report a first attempt and a null.

We fitted a difference-of-means vector Δe between two-shot and zero-shot prompts and added $\alpha\,\Delta e$ at every unroll. At $\alpha = 0$ the generation is unchanged in all 25 items. First-token accuracy moves from 0.000 to 0.080 while a norm-matched random control is indistinguishable throughout; over 132 conditions spanning both sides of the threshold, the largest $|\Delta R|$ is 0.0065 and no condition crosses it.

Scope. These are one intervention measured twice, not two independent nulls, and the dose was adequate — oracle accuracy degrades from 0.560 to 0.480 at large α, so the perturbation acted. We read this as the wrong injection site rather than as orthogonality of directions, the regime's onset being 33 positions upstream of where we applied the vector; applying Δe at all positions is the discriminating experiment and was not run. We banked $\|\Delta\|$ but not $\|e\|$ at the site, so α is not expressible as a fraction of scale.

4.6. Context costs depth rather than accuracy. Padding a bare prompt with irrelevant prose to the five-shot token length (length match verified, median ratio 1.000) collapses the gold's starting rank from 31 to 329, where five real exemplars leave it at 58.5. Oracle accuracy barely moves (0.500 to 0.479) while the depth at which the answer becomes available rises from 4.73 to 7.29 unrolls ($p = 0.0001$).

4.7. The generated prose is computation. Measuring the trajectory at every generated position rather than only the first, the forward pass that carries the answer is deeper than position 0 ($4.07 \to 7.89$, $p = 0.0031$) and deeper than the surrounding prose positions (7.89 against 3.65, $p = 0.0004$). The effect concentrates on arithmetic and is near-absent on copying.

§5. The Regime and the Answer Occupy Different Windows

The two preceding sections are connected by a timing fact that reframes both. The rotation statistic is computed over the tail of the trajectory; the answer, by contrast, is best-ranked at a median unroll of about four. Measuring the same statistic over the decision window (unrolls 0–11, the earliest a period-6 orbit is definable at all, since two cycles are required) and over the tail separates two facts that had been conflated. Over the decision window, rotating and settling prompts are indistinguishable: 0.229 against 0.230, a separation of 0.000. Over the tail the same prompts separate completely, 0.862 against 0.298. A sliding twelve-unroll window dates the divergence (Figure 1). Indexing each window by its start, the two curves are identical through the window beginning at unroll 12 (0.29 against 0.29) and first separate at the window beginning at unroll 16 (0.56 against 0.34); the rotating curve then crosses the regime threshold and saturates at 0.96, while the settling curve flattens at 0.55 and never reaches it.

The early window is not quiet. Its non-DC power exceeds the tail's by about two orders of magnitude (1.6×10^5 against 1943 and 59); it is energetic and simply carries no period-6 structure.

This changes what our own behavioural null means. We had found that a causally induced regime change alters almost nothing in what the model answers—one of eighteen paired units. That is not evidence that the regime is a dynamical epiphenomenon: the manipulation acted on a property that had not yet come into existence when the readout was settled. This is a timing claim rather than a causal one. It does not show that the regime could not affect behaviour, only that on this architecture the answer is fixed roughly twelve unrolls before the regime becomes measurable. It also bounds the instrument: no period-6 statistic can describe the state at the moment of decision, and we state this wherever the regime is connected to behaviour.

§6. The Three Starting Hypotheses

H1 (settle / loop / drift). Answered, in the form "both, at different levels". Unbounded drift is excluded by construction: every recorded state is a normalisation output and lies on a sphere of radius 76.386 (per-orbit relative deviation $\sim 5 \times 10^{-5}$), against which two random points on the same sphere lie 108.03 ± 0.75 apart — the correct null for any distance we quote. A large period-4 cycle exists across the four core blocks, its vertices separated by about half the state norm and its perimeter 1.6× the total distance travelled. Within a block the state settles or rotates according to the instruction token: on rotating orbits the per-block last-step residual is 2.9152 against 0.0185 for settling, a factor of 157, while the cycle's vertex geometry is unchanged between regimes.

H2 (harder problems recruit more depth). Architecturally untestable on this model. The hypothesis as the literature states it concerns per-position adaptive depth, which adaptive-computation models measure directly and confirm: ponder time grows with difficulty when a halting cost is trained in [2]; a universal transformer with dynamic halting reports mean per-symbol depth rising $2.3 \pm 0.8 \to 3.1 \pm 1.1 \to 3.8 \pm 2.2$ as the number of required supporting facts goes from one to three [4]; and the objective is stated as computation growing with problem complexity rather than input size [3]. Huginn has no halting mechanism: the whole sequence is unrolled together, so every position receives the same r and there is nothing to allocate. Depth does buy capability at fixed difficulty, but difficulty within a task moves recruited depth by +0.23 unrolls against task identity's +5.56. A difficulty axis often proposed for recurrence, sequential state tracking, is moreover out of reach for this architecture class on theoretical grounds: parity is the S_2 word problem and is argued to be unsolvable by these models, with many state-tracking problems harder than S_5 [5].

H3 (contraction implies no running register). Its strong form overstates its evidence. Contraction is real, but ρ is a family-level quantity rather than a model constant, spanning 0.8239 to 0.9181 across task families with a between-family spread some four times the within-family spread; the single value used for the timescale prediction above should be read as a representative rate, not a property of the model. The limit set retains ample room to separate states, and the running count is linearly decodable from the latents.

§7. Reliability: Controls, Retractions, and Instrument Nulls

Several results in this project were produced by instruments rather than by the model. We record them because the reliability of the remainder depends on it; Appendix C gives the full list.

A headline behavioural null was vacuous: the correctness variable it compared was false in 432 of 432 orbits, so there was no variance for the intervention to change; re-tested on an outcome that does vary, the original conclusion returned. A geometry headline was withdrawn by its own registered control within hours, and a spectral run was refused by its own consistency gate. A linear probe of the class used for several of our nulls cannot recover a label that is a guaranteed deterministic function of its own input features (0.690 against a 0.600 baseline, where the correct one-dimensional statistic reaches 0.980), so those nulls are uninformative rather than negative. And the extraction rule dominates the number: over 1424 banked generations, exact match recovers 0.038 of the answers and a last-number rule 0.240 at half the false-positive rate.

§8. Known Limitations

We do not reproduce the orbit reported for this model on a prompt of the kind its authors use to illustrate it. Sweeping every period the tail resolves, rather than the period-6 bin alone, finds no peak anywhere: the median non-DC power of that trajectory is 2.096, against 665–3139 for prompts we call rotating, with two further free-form prompts at 0.023 and 0.363. We therefore state the negative narrowly — numerical-reasoning content is not sufficient to produce the orbit we measure — noting that we ran a prompt in the shape of their example rather than their exact string, on 64 iterations against their 128. The phenomenon itself is theirs; it is the attribution to numerical content that our data contradict, on three independent grounds.

Our benchmark numbers and the published figure share an uncorrected surface-form bias inherent to likelihood scoring. The timing result says when the two properties become measurable, not whether one can influence the other; we state that hedge deliberately, since an intervention inside the decision window could still find an effect ours did not. Our direct behavioural test of H3 — injecting a donor state at h_0 — failed three times for reasons internal to our harness and is not reported; it remains the natural next experiment, though the authors' reported path independence over h_0 partly predicts a null. The authors also name a third limiting structure, "sliders", which we did not look for: our statistic is tuned to periodic structure and would not recognise one, so our labels read rotating-versus-not rather than as a taxonomy. Finally, comparisons to two related works are cross-checkpoint or cross-protocol and so are not directly numerically comparable.

§9. Conclusion

The recurrent map carries a sharp, causally controllable structure selected by the instruction's surface form rather than by the task it states. Our behavioural tests of that structure came back null, but the timing result shows why that is weak evidence: they intervened on a property not yet measurable when the answer was already fixed. Separately, much of what we and the standard protocols scored as capability was the model's willingness to emit its answer where a scorer looks. Whether the regime can affect behaviour remains open, and the experiment that would settle it must act earlier in the trajectory than ours did.

§10. Contribution

(To be completed by the authors — one entry per team member.) Arsenii Varaksin — to be completed. Alexander Shiianov — to be completed. David

Sverdlov — to be completed. Serguei Barannikov (mentor) — problem formulation, the hypotheses H1–H3 that this report tests, and supervision.

Acknowledgments

We thank the organisers of SMILES-26 for the school and for hosting this work, and Yandex Cloud for the compute and for assistance with it. The experiments reported here were run on Yandex DataSphere.

Appendix §A. Numerical Summary

Quantity	Value
Regime switch by one instruction word	36/36 rotating vs 0/36, token count fixed at 53
Semantic control	neighbours of symbol 0/144; of element 96/120
Form-matched control	72/72 rotating
Threshold reproduces binary label	691/696; 688/696 held out
Embedding-row edit flips regime	gates bit-exact at 0.000e+00
Regime by position	rotating prompts 62.0% of positions; settling 0.000
Mid-trajectory parameter swap	62/64 switches take
Relaxation asymmetry	entering 15.5 unrolls, leaving 1.0
Predicted from $\rho \approx 0.83$	16.1 unrolls
First-token vs produced answer	0.125 against 0.781
Chance rate for containment	0.042–0.152
The as opening token	39.4% of 1260 generations
Gold's worst final rank (feasible tasks)	35, over 2720 draws
ARC-Easy, published	0.699 at $r = 32$
ARC-Easy, matched protocol	0.658
ARC-Easy, five-shot letter argmax	0.723 ($p = 3.35\text{e}{-07}$)
Two-shot effect on oracle accuracy	+0.221 ($p = 9.3\text{e}{-09}$)
Irrelevant padding, start rank	31 $\rightarrow$ 329 ($p = 2.9\text{e}{-09}$)
Answer position vs prose positions	7.89 against 3.65 unrolls ($p = 0.0004$)

Table 1. Principal quantitative results. Inference-only on the pinned revision.

Appendix §B. Index of Experiments

Each result in the body traces to one run. Every run is inference-only on the pinned revision, with the initial state seeded before each forward pass so that repeated forwards are comparable.

§	Run	Design
3.1	word swap	18 sequences × 3 instructions × 2 markers, token count fixed at 53
3.1	noun sweep	16 nouns × 12 shared sequences × 2 markers (384 orbits)
3.1	threshold	re-analysis of 696 orbits over three banks, no GPU
3.2	embedding swap	252 orbits; interpolation of one wte row, both endpoints gated
3.3	position sweep	21 prompts, every token position, 64 unrolls
3.4	hysteresis	64 mid-trajectory parameter swaps at 8 switch points, both directions
4.1	generation scoring	128 generations at $r = 32$ and 48, with measured chance rates
4.2	depth bank	1260 generations, 21 families, 5 depths (re-analysis, no GPU)
4.3	benchmark arms	120 items, 4 scoring/prompt arms; plus a 101-item paired shot comparison
4.4	format sweep	24 items per arm, instruction wording varied at fixed content
4.4	demonstration	paired 0- vs 2- vs 5-shot on the census families
4.5	steering	132 conditions spanning both sides of the regime threshold
4.6	length control	zero-shot padded with neutral prose to the 5-shot token length
4.7	answer position	36 generations, every generated position, 48 unrolls
5	window split	48 prompts; the same statistic on the decision window and the tail
7	probe control	204 banked arrays over 51 nouns, leave-one-noun-out

Table 2. The experiments behind each section, with their designs.

Raw outputs for every row above are retained, so each number in the body can be recomputed without re-running the model.

Appendix §C. Withdrawn and Amended Claims

Ten ledger rows carry an explicit withdrawal or amendment marker. Every entry below states what fell and what survived, because in each case something did.

Inherited from earlier work. D34 and D36 are superseded by D50: the direction they measured is 97% the current-token contrast, so the window statistic they rest on is an artefact — the register claim itself survives, the stated mechanism does not. D43 is retracted; it held that the contraction rate predicts how deep the model can usefully think, and direct measurement flipped the verdict. D51 is subsumed by a same-day rerun on more checkpoints, though its caveat about the early jump was confirmed.

The regime line. D137/D138 are replaced by the threshold result: their "gapped, therefore two regimes" test used a uniform null, which any clustered distribution beats, so it never tested the claim being made; checked against the labels, the gap splits the settling population. The replacement is stronger, reproducing the binary label on 691 of 696 orbits. D135/D136 are superseded because the probe class they used cannot resolve the question (Section 7); what survives is five held-out replications. D164 was amended by its own registered norm control, which the decisive matched pair did not survive — the measurements replicate bit-for-bit, the interpretation is what failed.

The evaluation line. D148's correctness half is withdrawn as vacuous: correct was False in 432 of 432 orbits, so there was no variance for the regime to change. Re-tested on an outcome that varies, the conclusion returned at 1 of 18 paired units. D162's headline refuted "protocol" after varying only the scoring rule at fixed prompt; prompt format was the whole story and was never varied. D193's parsed-exact column measured our parser rather than the model, reading 14× low on the harness arm; on the repaired metric its conclusion survives unchanged.

Narrowed rather than withdrawn. Three interpretations were tightened without the underlying row falling: arithmetic families are not "uncertain about the answer" but compute in prose; depth tracks where the answer starts rather than the kind of operation; and the steering nulls are bounded to the last-position injection site rather than to orthogonality of directions.

Refused by their own gates. Two runs never became claims at all: a spectral pass whose tail length disagreed with the statistic it was validating against, and a reasoning arm that hit its wall-clock budget and banked cleanly without executing.

Appendix §D. A Companion Study: Step Normalisation in Two-Scale Latent Dynamics

This appendix reports a study carried out by David Sverdlov alongside the main investigation. It concerns a different object from the rest of the paper, and we present it separately for that reason rather than folding its numbers into the results above.

What it did. Pappone et al. [7] describe latent trajectories in recurrent-depth transformers as having two scales, with successive update steps tending towards orthogonality. This study asked whether that structure appears in a controlled setting where the dynamics can be manipulated directly. A small recurrent model was built for the purpose — 128 dimensions, an orthogonally initialised rotation layer, trained as an autoencoder for 50 epochs on one thousand FineWeb texts — and the angle between successive update steps was measured across depths 2 to 5, thirty-two steps per trajectory.

What it found. Successive steps anti-align rather than turning through a consistent small angle: the mean cosine between consecutive steps sits at -0.494 to -0.502 across the banked runs, an angle near $134°$, with the step norm decaying monotonically over the trajectory. The finding we consider useful is methodological. Before normalising the step, the measured orthogonality is confounded with the step norm, so the statistic partly tracks how far the state moved rather than in which direction; after normalisation the correlation between the two falls to -0.0436 and the angle becomes stable across the trajectory. Any measurement of step-to-step geometry that does not normalise first is reporting a mixture of the two quantities.

What it does not establish. It is not a measurement of Huginn-3.5B, and no number in it should be read as one. The model is separately trained and two orders of magnitude smaller in width; the banked trajectories it analyses are labelled synthetic; and because the rotation layer is orthogonally initialised, the presence of orthogonal structure is in part imposed by construction rather than discovered. The study therefore bears on how to measure consecutive-step geometry, not on what the checkpoint studied in the main body does.

Why it is here. The methodological point transfers directly, and it applies to a gap in our own work. Our rotation statistic is a frequency-domain quantity computed over a whole window; it says nothing about the direction of individual steps, and we never measured consecutive-step geometry on Huginn at all. Running this measurement on the trajectories analysed in Sections 3 and 5 — with the normalisation this study shows to be necessary — is a well-defined next experiment, and one for which the instrument now exists.

Appendix §E. Why a Statistic Rather Than a Projection

The published account of this model's latent dynamics is established by projecting trajectories onto their leading principal components and reading the pictures. That practice cannot support the claims we needed to make, for four reasons.

A projection is fitted to the data it judges. Two components of a 5280-dimensional trajectory retain a small fraction of the variance directions, and which fraction is chosen by the trajectory itself. Whether a loop appears is therefore a property of the basis as much as of the dynamics.

Damping is confounded with rotation. A monotone settle along a curved path renders as an arc, and an arc read as a partial orbit cannot be falsified by looking harder at the same picture.

A projection cannot support a negative. "This prompt does not orbit" does not follow from a picture in which no orbit is evident. Two of our central results are negatives — that the authors' own illustrative prompt does not orbit at any period the tail resolves, and that settling prompts rotate at none of their token positions — and neither is available by inspection.

A projection cannot support a threshold. Without one, "rotating" is a judgement made per figure: no population statement, no held-out validation, no cross-prompt comparison. The threshold $R > 0.6677$ is what licenses 691 of 696 and 688 held out.

An error of exactly this kind, which we made. The first pass of our period sweep computed the spectrum over a 48-unroll window while comparing against the 24-unroll statistic. It failed its own consistency gate and was voided. The cause is the damping confound above: over the longer window the decay transient dominates the low-frequency bins, so the most strongly rotating prompt peaked at period 48 — the slowest bin available — rather than at 6. For a damped orbit a longer window is worse, not better. Read as a picture, that error would have produced a plausible figure and no alarm; read as a number checked against its own definition, it produced a failed gate and a voided run.

What this does not claim. The phenomenon is not ours. The authors observed orbiting and named it, and our disagreement is with the attribution of it to numerical-reasoning content, not with its existence. What we add is an instrument with a threshold and a null, which makes the property measurable rather than illustrable.

References

1. Jonas Geiping et al., Scaling up test-time compute with latent reasoning: a recurrent depth approach, arXiv:2502.05171, 2025.
2. Alex Graves, Adaptive computation time for recurrent neural networks, arXiv:1603.08983, 2016.
3. Andrea Banino, Jan Balaguer, and Charles Blundell, PonderNet: learning to ponder, arXiv:2107.05407, 2021.
4. Mostafa Dehghani et al., Universal transformers, arXiv:1807.03819, 2018.
5. Riccardo Grazzi et al., Unlocking state-tracking in linear RNNs through negative eigenvalues, arXiv:2411.12537, 2024.
6. Shaojie Bai, J. Zico Kolter, and Vladlen Koltun, Deep equilibrium models, arXiv:1909.01377, 2019.
7. Francesco Pappone, Donato Crisostomi, and Emanuele Rodolà, Two-scale latent dynamics for recurrent-depth transformers, arXiv:2509.23314, 2025.

Что кодирует латентная траектория трансформера с рекуррентной глубиной

Арсений Вараксин, Александр Шиянов, Давид Свердлов
Научный руководитель: Сергей Баранников (Сколтех)

Аннотация

Исследуется латентная траектория модели Huginn-3.5B — трансформера с рекуррентной глубиной. Показано, что одно слово в инструкции переключает рекуррентное отображение между режимом сходимости и затухающим вращением с периодом 6, причём переключение причинно управляется одной строкой матрицы вложений. Показано также, что значительная часть измерений качества модели фактически измеряла формат ответа, а не его наличие. Приводятся опровергнутые и уточнённые утверждения.

Ключевые слова: трансформеры с рекуррентной глубиной, латентные рассуждения, интерпретируемость, протоколы оценки.